

Effect of Fiber Orientation on Buckling Behavior of Laminated Composite Plates Using Abaqus



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INTRODUCTION

Composite materials are widely used in aerospace, automotive, and structural applications due to:

- High strength-to-weight ratio
- High stiffness
- Tailorable mechanical properties

Thin composite structures may fail due to **buckling** under compressive loads. Therefore, studying buckling behavior is important.

PROBLEM STATEMENT

The buckling behavior of laminated composites depends strongly on:

- Fiber orientation
- Boundary conditions
- Applied loading

There is a need to identify which Fiber orientation provides better buckling resistance.

OBJECTIVES

The objectives of this work are:

1. To model laminated composite plates in Abaqus.
2. To perform **eigenvalue buckling** analysis.
3. To compare different fiber orientations.
4. To identify the optimum orientation.

GEOMETRY AND MATERIAL PROPERTIES

Geometry of Composite Plate

Parameter	Value
Length	130 mm
Width	40 mm
Thickness	1 mm

Laminate Configuration

- Number of plies = 8
- Single ply thickness = 0.125 mm
- Total laminate thickness = 1 mm

Material Description:

E-glass fiber (270 g/m²) and epoxy resin system (CY225 resin + HY225 hardener).

Property	E_1 (MPa)	$E_2 = E_3$ (MPa)	$G_{12} = G_{13} = G_{23}$ (MPa)	$\nu_{12} = \nu_{13} = \nu_{23}$
Value	44150	12300	4092	20

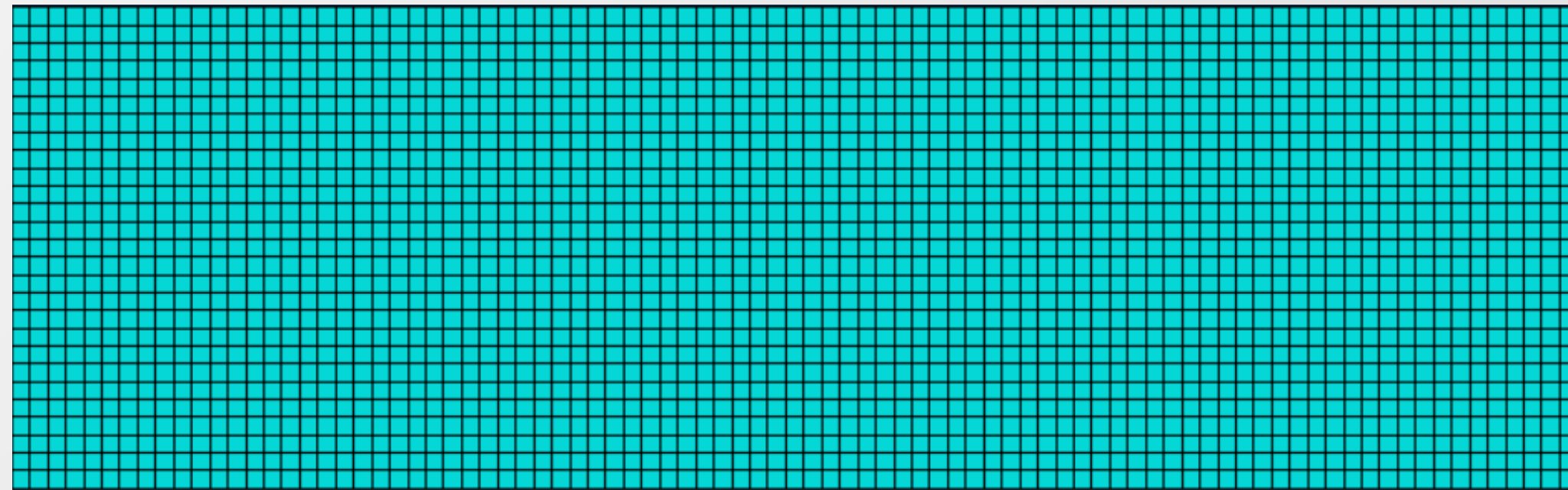
ORIENTATION AND MESHING

The following laminates were analyzed:

- [0/0/0]
- [45/45/45]
- [90/90/90]
- [0/45/0]
- [0/45/90]
- [45/90/45]
- [-45/45/-45]

Mesh Details

- Element type: Shell element
- Mesh size: 1.5 mm
- Uniform structured mesh was used throughout the model.



A mesh size of 1.5 mm was selected to ensure adequate geometric discretization and accurate buckling mode prediction.

BOUNDARY CONDITIONS AND LOADING

Left Upper Corner Node

- $U_x=0$
- $U_y=0$
- $U_z=0$

This node acts as the reference point to prevent rigid body motion.

Left Edge

- $U_x=0$
- $U_z=0$

This allows deformation in the longitudinal direction while restricting out-of-plane displacement.

Right Edge

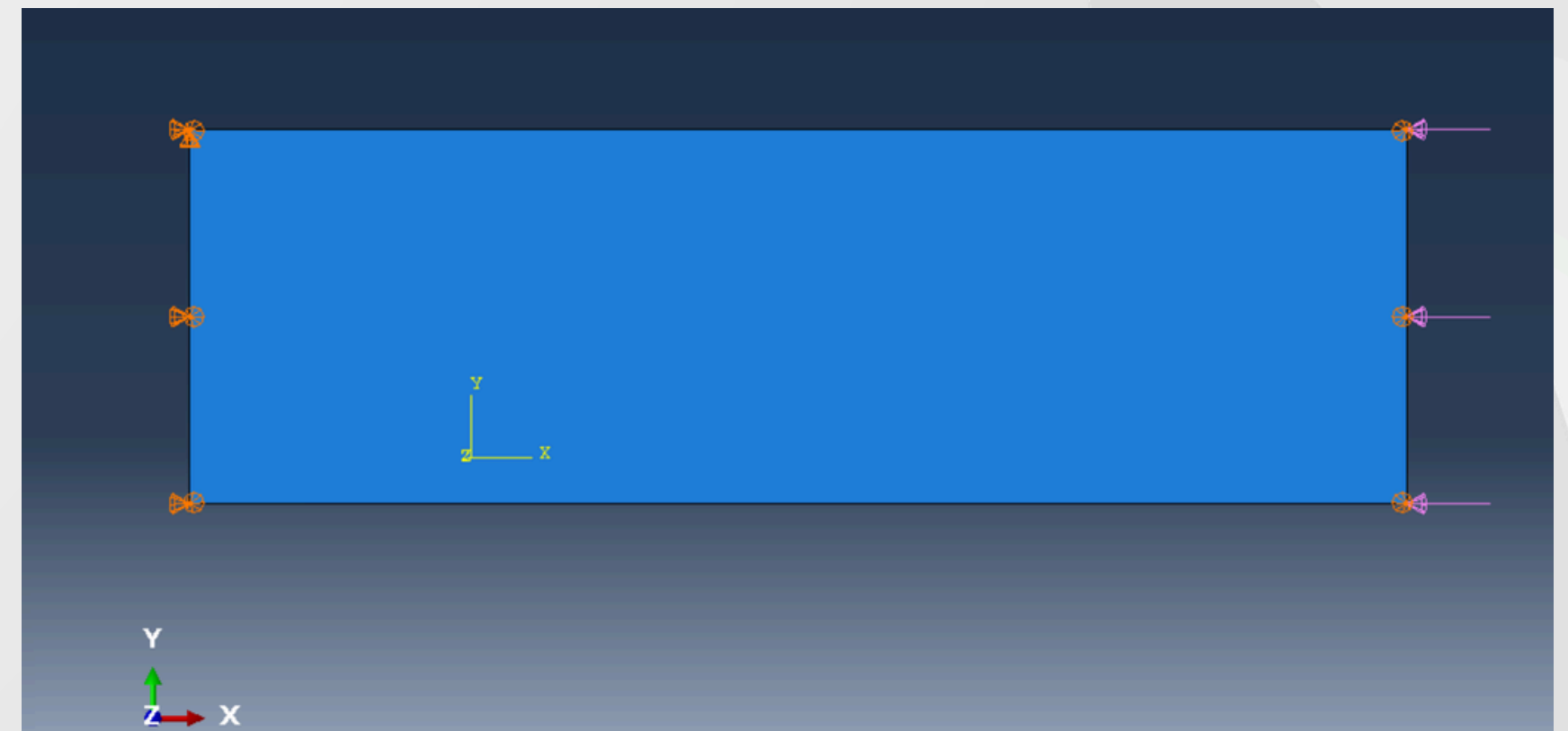
The loaded edge was constrained in the thickness direction:

- $U_z=0$

This ensures plate stability during compressive loading.

Loading Condition

A compressive pressure of 1.5 MPa was applied on the right edge of the composite plate.



FINITE ELEMENT ANALYSIS

Buckling analysis was performed using eigenvalue extraction.

The governing eigenvalue equation is:

$$[K+\lambda S]\psi=0$$

Where:

- K = stiffness matrix
- S = stress stiffness matrix
- λ = buckling load factor

λ = Buckling Load/ Applied Load

This method determines critical buckling loads and mode shapes.

FINITE ELEMENT ANALYSIS

Applied pressure = 1.5 MPa = 1.5 N/mm²

Plate dimensions:

- Length = 130 mm
- Width = 40 mm
- Thickness = 1 mm

Since pressure is applied on the loaded edge, edge loaded area:

$$A = \text{width} \times \text{thickness}$$

$$A = 40 \times 1 = 40 \text{ mm}^2$$

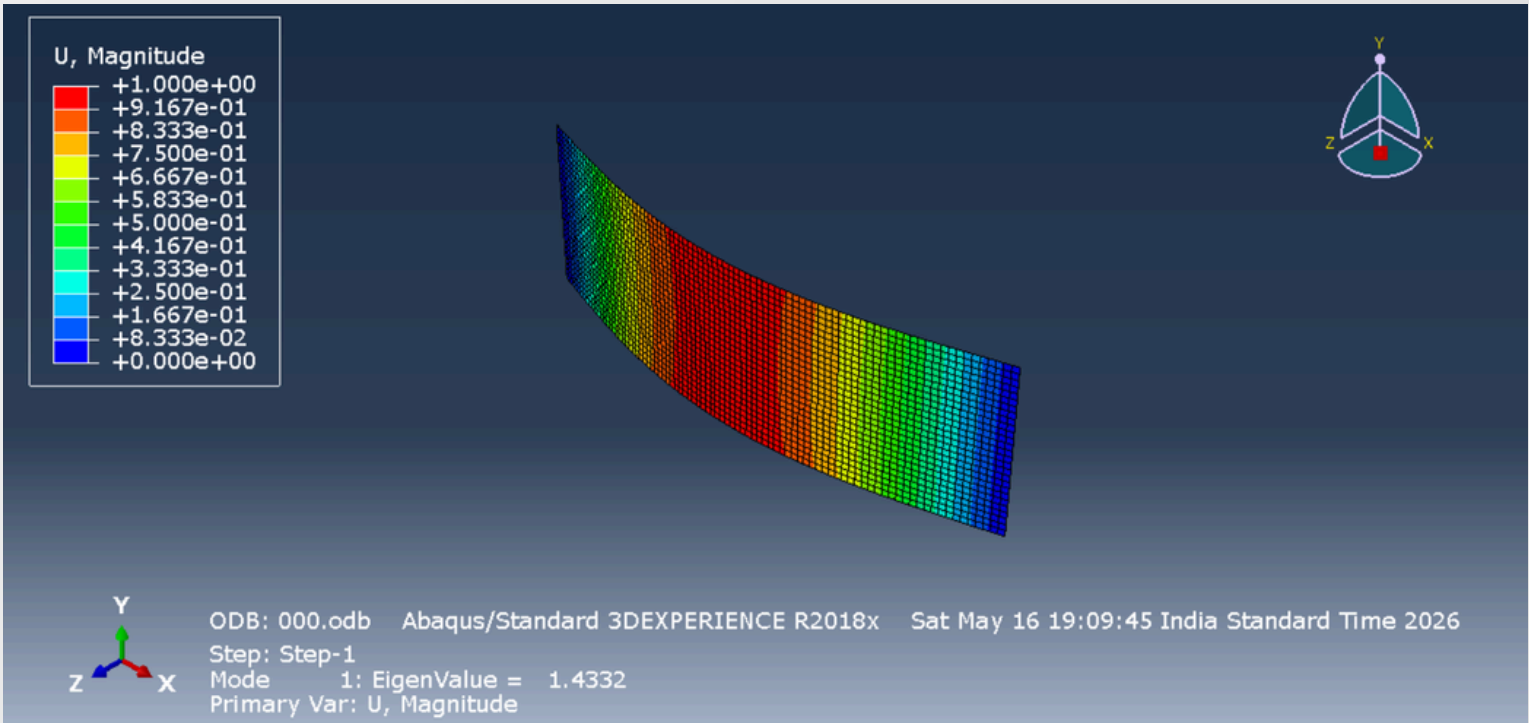
Applied load:

$$P(\text{applied}) = 1.5 \times 40 = 60 \text{ N}$$

$$P_{cr} = \lambda \times P(\text{applied})$$

RESULT

Orientation	Eigenvalue (λ)	Critical Buckling Load (Pcr)
[0/0/0]	14232×10^{-4}	85.39 N
[45/45/45]	45885×10^{-5}	27.53 N
[90/90/90]	39920×10^{-5}	23.95 N
[0/45/0]	43667×10^{-5}	26.20 N
[0/45/90]	39516×10^{-5}	23.71 N
[45/90/45]	46382×10^{-5}	27.83 N
[-45/45/-45]	45028×10^{-5}	27.02 N



DISCUSSION OF RESULTS

Effect of fiber orientation

The [0/0/0] showed the highest critical buckling load:

$$P_{cr}=85.39N$$

This indicates maximum axial stiffness and better resistance against compressive instability.

Lowest buckling resistance

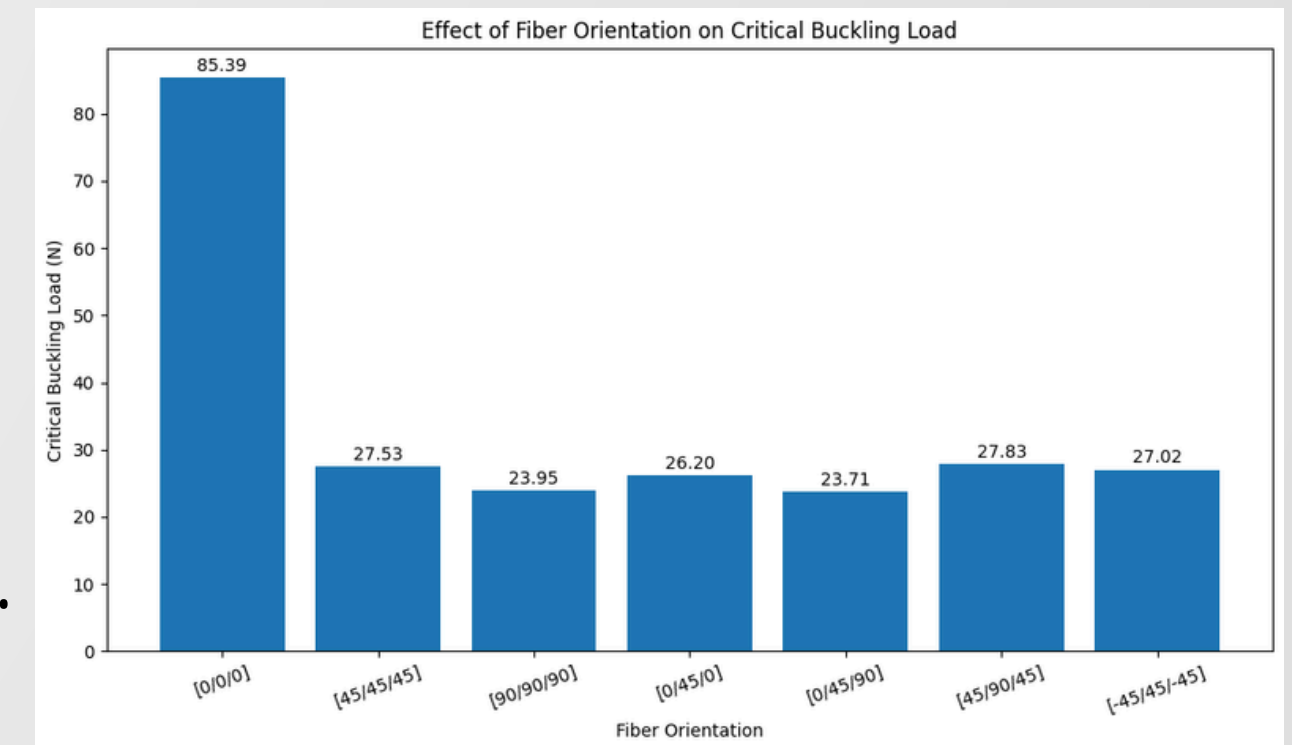
The [0/45/90] laminate showed the minimum buckling load:

$$P_{cr}=23.71N$$

This suggests reduced effective stiffness due to mixed fiber orientations.

Mode shape behavior

The first buckling mode governs structural instability and is critical for design evaluation.



CONCLUSION

1. Fiber orientation has a **significant effect** on buckling resistance.
2. The [0/0/0] laminate showed the highest critical buckling load (85.39 N) and therefore provides the 3. best buckling performance under compressive loading.
4. The [0/45/90] laminate showed the lowest critical buckling load (23.71 N), indicating lower structural stability.
5. Angle-ply laminates showed moderate buckling performance compared to unidirectional laminates.
6. Eigenvalue buckling analysis in Abaqus successfully **predicted critical buckling** loads and mode shapes for different stacking sequences.

REFERENCES

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THANK YOU